

Vacuum as Minimal Concrete Persistence Regime

The Non-Null Baseline Role of Field-Carrier Structure

Amos Jay Maley

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Abstract

This paper derives the minimal vacuum role within the realized-physics arc. Previous results established that realized persistence cannot be physically null, that standing-bearing persistence localizes, that localized standing induces continuation-neighborhood domains, that stable comparison supplies the metric role of a quotient comparison carrier, that perturbation transport is bounded by continuation-comparison constraints, that fields are persistence carriers, that particles are localized persistence classes, that gauge structure is boundary-fixed presentation equivalence, and that mass and inertia are deformation-response structure. The present paper asks what remains when localized particle-class excitations are absent. It shows that the resulting regime cannot collapse into total vacuity, empty-container spacetime, field absence, trace-only bookkeeping, or arbitrary ground-state selection. A physical vacuum must retain the minimal carrier, boundary, transport, comparison, gauge, restoration, and deformation-response roles required for concrete persistence.

The main result is a vacuum-role induction theorem. In a fixed admissible realized persistence regime with field-carrier mediation Φ_D , bounded perturbation transport, quotient comparison carrier C_D , particle persistence classes Π_D , gauge-role structure $\mathcal{G}_D$, and mass/inertia role $\mathcal{M}_D$, the absence of localized particle-class excitations under minimal concrete persistence induces a non-null vacuum profile

$$\Omega_D(U) = (\Phi_U^0, \partial_U^0, T_U^0, Q_U^0, \mathcal{G}_U^0, \mu_U^0, \mathcal{R}_U^0).$$

This profile defines the minimal physical vacuum role: no-particle relative to localized persistence classes, yet non-null relative to field-carrier and persistence structure. Fock vacua, Hamiltonian ground states, zero-point energy, Higgs vacuum expectation values, condensates, cosmological constants, dark-energy terms, stress-energy assignments, and vacuum sectors of field theories are not assumed. They are downstream specializations of the non-null baseline persistence role isolated here.

1 Introduction

The realized-physics arc has proceeded by deriving familiar physical roles from standing-bearing persistence rather than importing them as primitive. The first paper establishes the non-null floor of physical realization: realized persistence cannot be reconstruction-null inside its own admissible continuation regime [1]. The second derives bounded standing-localization as the origin of physical locality [2]. The third derives continuation-neighborhood domains as the origin of local-domain structure [3]. The fourth derives the metric role as quotient-stable comparative representation among local continuation domains [4]. The fifth derives bounded perturbation transport as the finite-propagation-constraint role [5]. The sixth derives fields as persistence

carriers [6]. The seventh derives particles as localized persistence classes [7]. The eighth derives gauge structure as boundary-trace-fixed presentation equivalence [8]. The ninth derives mass and inertia as deformation-response structure [9].

In compact form, the chain is

$$\begin{aligned} \text{RealPers}_D &\Rightarrow \text{NonVacRec}_D \Rightarrow [B_D]_{\equiv_B} \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \\ &\Rightarrow C_D \Rightarrow \text{BoundTrans}_D \Rightarrow \Phi_D \Rightarrow \Pi_D \Rightarrow \mathcal{G}_D \Rightarrow \mathcal{M}_D. \end{aligned}$$

The present paper asks the natural next question: what remains when localized particle persistence classes are absent? The answer is not nothing. Removing localized particle-class excitations removes Π_D -level excitations above a baseline. It does not remove field-carrier mediation, boundary traces, quotient comparison, bounded transport possibility, gauge-compatible presentation, restoration capacity, or deformation-response structure. The physical vacuum is therefore the minimal non-null concrete persistence baseline relative to which particles, transport, mass/inertia, gauge structure, and excitations become meaningful.

1.1 Theorem claim and physical role

The theorem-level claim is

$$(\Phi_D, \text{BoundTrans}_D, C_D, \Pi_D, \mathcal{G}_D, \mathcal{M}_D) \wedge \text{MinPersist}_D \wedge \text{NoPartExc}_D \Rightarrow \text{VacRole}_D.$$

The physical-role claim is that VacRole_D is the minimal vacuum role occupied by later ground states, no-particle sectors, zero-point structures, order-parameter baselines, and cosmological vacuum terms. The paper does not treat vacuum as primitive empty space, as a bookkeeping zero, or as a null domain. It treats vacuum as the minimal concrete persistence regime that remains when localized particle-class excitations are absent.

This distinction is essential. A no-particle condition is relative to particle persistence classes; it is not a nullity condition. A null regime lacks the carrier, boundary, transport, comparison, gauge, restoration, and deformation-response structures required for physical realization. A vacuum regime retains those structures in their minimal baseline form.

Paper	Structural result	Physical role
1	NonVacRec_D	physical realization cannot be null
2	$[B_D]_{\equiv_B}$	origin of physical locality
3	$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D)$	origin of local-domain organization
4	$C_D = \text{CompWit}_D / \sim_D$	origin of metric comparison role
5	BoundTrans_D	origin of finite-propagation constraint
6	Φ_D	origin of field-carrier mediation
7	$\Pi_D = [\kappa_U]_{\equiv_\Pi}$	origin of particle identity
8	$\mathcal{G}_D$	origin of gauge structure
9	$\mathcal{M}_D = (\mu_D, \iota_D)$	origin of mass/inertia role
10	Ω_D	origin of vacuum structure

1.2 Scope

The paper derives the minimal vacuum role. It does not derive a Fock vacuum, Hamiltonian ground state, zero-point energy value, Higgs vacuum expectation value, condensate, false vacuum,

vacuum decay channel, vacuum polarization, Casimir effect, de Sitter vacuum, stress-energy tensor, cosmological constant, dark-energy model, or full vacuum sector of quantum field theory. Those are later specializations. What is derived here is the prior physical role: no localized particle persistence class above a non-null carrier baseline.

2 Dependencies and imported structure

Paper 10 uses all prior layers of the realized-physics arc. Its distinctive role is to distinguish particle absence from physical nullity after fields, particles, gauge structure, and mass/inertia have already been reconstructed. The dependency order is summarized in Table 1.

Table 1: Imported structure for the vacuum-role derivation.

Imported result	Role in the present paper
Non-vacuous realized persistence	Vacuum cannot be total nullity; it must retain concrete persistence structure.
Bounded standing-localization	Vacuum baseline cannot be diffuse global standing without bounded local role structure.
Continuation-neighborhood domains	Vacuum is organized over local continuation domains rather than bare empty extension.
Quotient comparison carrier C_D	Vacuum comparisons and baselines require quotient-stable comparison, not primitive scale choice.
Bounded perturbation transport	Vacuum retains transport-possibility structure; it is not pathless nothingness.
Field-carrier mediation Φ_D	Vacuum is baseline field-carrier structure, not absence of field-carrier mediation.
Particle persistence classes Π_D	Vacuum is absence of localized particle-class excitations above baseline, not absence of persistence.
Gauge-role structure $\mathcal{G}_D$	Vacuum must be gauge-compatible; gauge presentation cannot create or destroy vacuum role.
Mass/inertia role $\mathcal{M}_D$	Vacuum retains a deformation-response baseline relative to which mass/inertia profiles are defined.
Null-regime and concrete-domain closure	Vacuum cannot be trace-only, skin-only, abstract-only, or bookkeeping-only.
AMetric boundary	No primitive empty metric container, default ground-state representative, or hidden vacuum selector is admissible.
Vacuum/dark-sector sources	Effective vacuum energy and cosmological residuals are downstream extractions, not primitive vacuum ontology.

The structural continuity with Paper 1 is especially important. Paper 1 excludes physically null realization. Paper 10 identifies the vacuum as the minimal non-null concrete regime remaining after localized particle excitations are removed. In this sense, Paper 10 closes the first physical loop of the arc: vacuum is not a return to the null regime, but the baseline physical realization compatible with the structures derived in Papers 1–9.

3 Definitions and type discipline

Definition 3.1 (Field-carrier baseline). A field-carrier baseline over a local continuation domain U is the minimal carrier profile Φ_U^0 that remains when localized particle-class excitations are absent. It preserves the local carrier role required for continuation, transport possibility, boundary compatibility, and restoration, but contains no localized particle persistence class above the baseline.

Definition 3.2 (Particle-class excitation). A particle-class excitation is a localized carrier-mediated configuration that induces a particle persistence class

$$\Pi_D = [\kappa_U]_{\equiv \Pi}$$

above a baseline vacuum profile. The term “excitation” is used role-theoretically here: it denotes a localized persistence class relative to a baseline, not a Fock-space operator excitation or a completed quantum field-theoretic construction.

Definition 3.3 (No-particle condition). For a vacuum candidate Ω_D ,

$$\text{NoPartExc}_D(\Omega_D)$$

means that no localized carrier configuration in Ω_D induces a particle persistence class above the vacuum baseline. This condition does not imply $\Omega_D = \emptyset$. It removes particle-class excitations, not the field-carrier baseline.

Definition 3.4 (Minimal concrete persistence regime). A regime Ω_D satisfies $\text{MinPersist}_D(\Omega_D)$ iff it preserves exactly the carrier, boundary, transport, quotient comparison, gauge, restoration, and deformation-response roles required for concrete realized persistence, while containing no removable surplus particle-class excitation. Minimality is role-minimality, not absence of structure.

Definition 3.5 (Vacuum profile). A vacuum profile is a partial local-domain assignment

$$\Omega_D : \text{Dom}(\Omega_D) \subseteq \text{LCD}_D \rightarrow \text{VacProf}_D$$

with local components

$$\Omega_D(U) = (\Phi_U^0, \partial_U^0, T_U^0, Q_U^0, \mathcal{G}_U^0, \mu_U^0, \mathcal{R}_U^0).$$

Here Φ_U^0 is the baseline field-carrier profile, ∂_U^0 the boundary-trace interface, T_U^0 the transport-possibility interface, Q_U^0 the quotient comparison baseline, $\mathcal{G}_U^0$ the gauge-compatible presentation baseline, μ_U^0 the deformation-response baseline, and $\mathcal{R}_U^0$ the restoration/persistence baseline.

Definition 3.6 (Vacuum role).

$$\text{VacRole}_D$$

is the minimal physical vacuum role realized by such an Ω_D : a non-null, concrete, no-particle baseline preserving the structure required for realized persistence and for later field, particle, gauge, mass/inertia, and transport specializations.

4 Why vacuum cannot be null

A physical vacuum cannot be identified with the null regime. The null regime lacks precisely the structures that make vacuum physically meaningful: carrier baseline, boundary interface, transport possibility, quotient comparison, gauge-compatible presentation, deformation-response baseline, and restoration capacity. The vacuum is minimal, not empty of all physical role.

Theorem 4.1 (No null vacuum). *In a fixed admissible realized persistence regime, a physical vacuum role cannot be total vacuity. If VacRole_D is realized, then $\Omega_D \neq \emptyset$.*

Proof. Assume that a vacuum role is total vacuity. Then it lacks field-carrier mediation, boundary trace, quotient comparison, bounded transport possibility, gauge-compatible presentation, deformation-response baseline, and restoration structure. Without those structures, the candidate cannot support realized persistence, cannot define no-particle relative to particle classes, cannot distinguish baseline carrier structure from excitation, and cannot supply the conditions required for later physical specialization. Such a candidate is the null regime excluded by non-vacuous realized persistence and concrete-domain closure. Hence a realized physical vacuum must be non-null. $\square$

Proposition 4.2 (No-particle/non-null distinction). *The absence of localized particle persistence classes removes particle-class excitations; it does not remove the vacuum baseline:*

$$\text{NoPartExc}_D(\Omega_D) \not\Rightarrow \text{Null}_D.$$

Proof. $\text{NoPartExc}_D(\Omega_D)$ is defined relative to the particle-role layer Π_D . It states that no localized carrier configuration induces a particle persistence class above the baseline. It says nothing about the absence of field-carrier mediation, boundary trace, transport possibility, quotient comparison, gauge compatibility, restoration, or deformation-response baseline. Therefore no-particle is a relative excitation condition, not a nullity condition. $\square$

This distinction is the central physical content of the paper. Vacuum is no-particle relative to Π_D , but non-null relative to Φ_D , C_D , BoundTrans_D , $\mathcal{G}_D$, and $\mathcal{M}_D$.

5 Failure of empty-vacuum candidates

The familiar language of emptiness is inadequate unless it is carefully typed. Empty spacetime, field absence, trace-only zero, or primitive background selection do not yield a vacuum role. They either fail to support physical persistence or covertly import the very structures they claim to remove.

Lemma 5.1 (Empty-container insufficiency). *An empty spacetime container cannot serve as the vacuum role unless it preserves the carrier, boundary, transport, gauge, comparison, restoration, and deformation-response roles required for concrete persistence.*

Proof. If the container lacks those roles, it is physically inert with respect to the realized persistence regime: it cannot support transport, excitation, gauge-compatible presentation, deformation response, or local-domain continuity. If it contains those roles, it is not an empty container but a structured vacuum baseline. Thus the physical vacuum is not bare empty extension. $\square$

Lemma 5.2 (Field-absence collapse). *If field-carrier mediation is absent, then bounded perturbation transport, particle persistence classes, gauge role, and deformation-response structure cannot be supported. Therefore vacuum cannot be absence of field-carrier structure.*

Proof. Papers 6–9 derive field-carrier mediation as the support of localized persistence, particle identity, gauge-compatible presentation, and mass/inertia response. Removing field-carrier mediation removes the support for all of those roles. That gives null or physically inert absence, not vacuum. A vacuum must retain baseline field-carrier mediation Φ_D^0 . $\square$

Lemma 5.3 (Trace-only vacuum collapse). *A vacuum label, zero entry, coordinate notation, or empty-sector name carries no vacuum role unless it preserves the field-carrier baseline, boundary interface, transport possibility, quotient comparison, gauge compatibility, restoration, and deformation response of the regime.*

Proof. A trace-only assignment changes the description without altering the persistence-bearing organization of the regime. Such a label is skin. If it carries the structures required for vacuum role, then it is no longer trace-only; it is a representation of Ω_D . Hence trace-only vacancy is insufficient. $\square$

6 Positive vacuum-role requirements

The vacuum role is positive, not merely negative. It contains exactly the minimal baseline structures required for concrete persistence after particle-class excitations are absent.

Proposition 6.1 (Baseline carrier requirement). *A physical vacuum role requires a baseline carrier profile Φ_D^0 over local continuation domains. This profile preserves boundary trace, transport possibility, quotient comparison, gauge-compatible presentation, restoration profile, and deformation-response baseline.*

Proof. Without a baseline carrier profile, no local continuation domain can support realized persistence after particle-class excitations are absent. The result would be field absence, which collapses the roles derived in Papers 5–9. Hence Φ_D^0 is required. $\square$

Proposition 6.2 (Gauge-compatible vacuum baseline). *A vacuum baseline must be invariant under admissible gauge-role presentation changes. If $\rho_U \sim_G \rho'_U$, then the corresponding vacuum profiles must preserve the same vacuum role:*

$$\Omega_D(U) \sim_{\Omega} \Omega'_D(U).$$

If a claimed vacuum distinction changes only under gauge-equivalent presentation, it is a presentation artifact.

Proof. Gauge structure preserves field-carrier mediation and particle persistence-class identity across local presentations. A vacuum baseline is the minimal carrier baseline for those structures. If gauge-equivalent presentations changed the physical vacuum role, then gauge presentation would create or destroy standing-bearing baseline content. That contradicts boundary-fixed gauge equivalence. Therefore the vacuum baseline must be gauge-compatible. $\square$

Proposition 6.3 (Vacuum deformation baseline). *A vacuum role requires a deformation-response baseline μ_D^0 . This baseline does not imply nonzero particle mass. It supplies the response structure relative to which particle mass/inertia profiles, neutral response cases, and massless classes are defined.*

Proof. Paper 9 defines mass/inertia roles as deformation-response profiles of localized particle persistence classes. Such profiles are meaningful only relative to a baseline response structure of the carrier regime. If no baseline exists, then deformation response cannot be distinguished from nullity or arbitrary scalar assignment. Hence a vacuum profile includes μ_D^0 , without requiring nonzero particle mass. $\square$

Proposition 6.4 (Vacuum neutrality and non-nullity). *Vacuum may be neutral relative to particle-class excitation while remaining non-null in carrier structure:*

$$\text{NoPartExc}_D(\Omega_D) \wedge \Omega_D \neq \emptyset.$$

Proof. Neutrality concerns the absence of particle-class excitations above the baseline. Non-nullity concerns the persistence-bearing baseline itself. These are different typed conditions. A vacuum can have no particle persistence classes while retaining field-carrier, boundary, transport, comparison, gauge, restoration, and deformation baseline structures. $\square$

7 Vacuum-role induction

Theorem 7.1 (Vacuum-role induction). *In a fixed admissible realized persistence regime with field-carrier mediation, bounded perturbation transport, quotient comparison, particle classes, gauge role, and mass/inertia role, the absence of localized particle-class excitations under minimal concrete persistence induces a vacuum role:*

$$(\Phi_D, \text{BoundTrans}_D, C_D, \Pi_D, \mathcal{G}_D, \mathcal{M}_D) \wedge \text{MinPersist}_D \wedge \text{NoPartExc}_D \Rightarrow \text{VacRole}_D.$$

Proof. Paper 1 excludes null physical realization. Papers 2–5 supply localization, local continuation domains, quotient comparison, and bounded transport. Paper 6 supplies field-carrier mediation. Paper 7 distinguishes localized particle persistence classes from the carrier regime that supports them. Paper 8 requires gauge-compatible presentation. Paper 9 supplies deformation-response baseline structure. When localized particle-class excitations are absent, the Π_D -level excitations are removed, but the carrier, boundary, transport, quotient, gauge, restoration, and deformation structures required for concrete persistence remain. Minimal concrete persistence removes surplus particle-class excitation while retaining the baseline roles necessary for realized persistence. The remaining non-null baseline is Ω_D , and hence the vacuum role is induced. $\square$

This theorem is the positive counterpart of Paper 1. Paper 1 says physical realization cannot collapse into nullity. Paper 10 says that the no-particle physical baseline is the minimal non-null realization regime left after localized particle-class excitations are absent.

8 No primitive vacuum-energy assignment

The vacuum role should not be confused with a primitive scalar energy assignment. Vacuum-energy values, zero-point extractions, cosmological constants, and dark-energy terms require additional ledgers: metric normalization, stress-energy extraction, renormalization prescription, model background, source split, and observational calibration. They are not the primitive definition of vacuum.

Theorem 8.1 (No primitive vacuum-energy assignment). *A vacuum-energy scalar or cosmological-constant term cannot be imported before the vacuum role and extraction ledger are fixed. If a value preserves the vacuum role under admissible extraction, it is a later representation. If not, it is selector import, bookkeeping, or presentation artifact.*

Proof. A scalar vacuum-energy assignment by itself does not define field-carrier baseline, boundary trace, transport possibility, quotient comparison, gauge compatibility, restoration, or deformation response. Therefore it cannot constitute the vacuum role. If the scalar is tied to an admissible extraction from Ω_D after the relevant metric, stress-energy, and model ledgers are fixed, then it may represent a later specialization of the vacuum role. Otherwise it is an imported number with no persistence-bearing authority inside the regime. $\square$

Corollary 8.2 (Hidden vacuum reservoir exclusion). *A latent vacuum structure inaccessible to admissible boundary traces, gauge-compatible presentation, transport witnesses, quotient comparison, and deformation-response baselines cannot carry realized vacuum role. If it becomes accessible, it belongs to Ω_D ; if it remains inaccessible, it is realization-inert.*

9 Relations to standard vacuum structures

9.1 Zero-point energy and vacuum fluctuations

Zero-point energy, vacuum fluctuations, and ground-state expectation values are later specializations of the non-null vacuum role. The present result does not compute them. It derives why a no-particle physical regime can still be non-empty in carrier, boundary, transport, comparison, gauge, restoration, and deformation-response structure. This is the prior role that zero-point descriptions may represent in field-theoretic settings [18, 19].

9.2 Higgs vacuum, condensates, and order parameters

Higgs vacuum expectation values, condensates, spontaneous-symmetry-breaking vacua, and order parameters are not assumed. They are ways the vacuum role may acquire additional carrier/gauge/deformation structure in later specializations. Paper 10 derives the baseline physical role: a minimal concrete persistence regime with no localized particle-class excitation above it. Higgs-sector and condensate structures would be enrichments of $(\Phi^0, \mathcal{G}^0, \mu^0)$, not replacements for the vacuum role.

9.3 Dark energy and the cosmological constant

An effective cosmological constant or dark-energy term is not primitive vacuum ontology. It is an extraction inside a metric/cosmological model after background choice, calibration, stress-energy split, and geometry/source assignment are fixed [22, 23]. The present result does not identify vacuum with dark energy. It supplies the non-null vacuum role that such later terms may represent if the relevant extraction ledger is fixed.

9.4 QFT vacuum sectors

Standard quantum field theory vacua may involve no-particle Fock states, ground states, operator-algebraic representations, symmetry-breaking sectors, superselection structure, and renormalization choices [20, 18]. Paper 10 derives the prior structural role common to these cases: no-particle

relative to localized persistence classes, but non-null relative to field-carrier baseline. No Fock-space formalism or operator vacuum is assumed here.

9.5 Particles and excitations

Particle classes Π_D are localized persistence classes above a vacuum baseline. Thus

$$\Pi_D \text{ is excitation-like relative to } \Omega_D.$$

This excitation relation is role-theoretic at this stage. It does not yet assert creation and annihilation operators, Fock space, or scattering dynamics. It says only that particles are localized persistence classes supported by and distinguishable from a non-null carrier baseline.

10 Finite witness model

Consider a finite witness model

$$G_D = (V, E, \Omega, \Phi, \Pi, \mathcal{G}, \mu),$$

where V are local continuation domains, E are admitted transport/adjacency relations, $\Omega(v)$ is the vacuum baseline profile, $\Phi(v)$ is the field-carrier profile, $\Pi(v)$ is a localized particle class when excitation is present, $\mathcal{G}$ is the gauge role, and μ is the deformation-response baseline. The witness graph instantiates one admissible realization of a vacuum baseline; it does not define vacuum as graph data.

The collapse cases are instructive:

- (i) If $\Omega = \emptyset$, the candidate is the null regime and cannot support realized persistence.
- (ii) If Φ^0 is absent, field-carrier baseline is absent and transport, particle, gauge, and deformation roles collapse.
- (iii) If boundary or transport interfaces are absent, concrete persistence is unsupported.
- (iv) If particle absence alone is imposed while $\Omega \neq \emptyset$, the model realizes a valid no-particle condition rather than nullity.
- (v) If a vacuum-energy scalar is assigned before Ω and an extraction ledger are fixed, the scalar is selector import.
- (vi) If a vacuum distinction changes only under gauge-equivalent presentation, the distinction is a presentation artifact.
- (vii) If a non-null minimal Ω preserves $(\Phi^0, \partial^0, T^0, Q^0, \mathcal{G}^0, \mu^0, \mathcal{R}^0)$, the model instantiates vacuum role.

11 Route exhaustion

Every proposed source of vacuum role falls into one of the following classes:

- (1) minimal concrete persistence baseline;

- (2) null regime;
- (3) particle absence alone;
- (4) empty spacetime container;
- (5) field absence;
- (6) primitive vacuum-energy scalar;
- (7) gauge-presentation artifact;
- (8) hidden vacuum reservoir;
- (9) measurement or extraction record alone;
- (10) cosmological model term imported before role fixation.

Cases (2)–(10) fail as primitive vacuum sources. Nullity lacks realized persistence. Particle absence alone removes excitations but does not specify a baseline. Empty container and field absence fail to support carrier, transport, gauge, and deformation roles. A primitive energy scalar or cosmological term requires an extraction ledger. Gauge presentation artifacts do not alter physical vacuum role. Hidden reservoirs are inert unless made admissibly accessible. Records and model terms are downstream fixations rather than vacuum-role sources. Only (1), the minimal concrete persistence baseline, survives.

Theorem 11.1 (Vacuum route exhaustion). *The only admissible source of physical vacuum role in the derived regime is minimal non-null concrete persistence baseline:*

$$\text{VacRole}_D \equiv \exists \Omega_D : \text{Dom}(\Omega_D) \subseteq \text{LCD}_D \rightarrow \text{VacProf}_D.$$

Proof. The preceding cases exhaust the possible readings of vacuum as nullity, no-particle condition, empty container, field absence, primitive energy, presentation artifact, hidden reservoir, extraction record, model term, or concrete baseline. All non-baseline candidates either lack the structures required for realized persistence, collapse into nullity, import selectors, or become downstream representations of an already-fixed baseline. Therefore the minimal concrete persistence baseline is the unique surviving vacuum-role source. $\square$

12 Normal form

The result of the paper can be written as

$$\boxed{(\Phi_D, \text{BoundTrans}_D, C_D, \Pi_D, \mathcal{G}_D, \mathcal{M}_D) \wedge \text{MinPersist}_D \wedge \text{NoPartExc}_D \Rightarrow \text{VacRole}_D}$$

with

$$\text{VacRole}_D \equiv \exists \Omega_D : \text{Dom}(\Omega_D) \subseteq \text{LCD}_D \rightarrow \text{VacProf}_D$$

and

$$\Omega_D(U) = (\Phi_U^0, \partial_U^0, T_U^0, Q_U^0, \mathcal{G}_U^0, \mu_U^0, \mathcal{R}_U^0).$$

The first line states that vacuum is induced only after field-carrier mediation, bounded transport, quotient comparison, particle classes, gauge role, and mass/inertia role are already available. The second and third lines state that vacuum is not nullity: it is a local-domain baseline assignment carrying the minimal structures required for concrete persistence.

13 Conclusion

Paper 1 established that realized persistence cannot be physically null. Paper 2 established bounded localization. Paper 3 derived continuation-neighborhood domains. Paper 4 derived the metric role of quotient-stable comparison. Paper 5 derived bounded perturbation transport. Paper 6 derived fields as persistence carriers. Paper 7 derived particles as localized persistence classes. Paper 8 derived gauge structure as boundary-fixed presentation equivalence. Paper 9 derived mass and inertia as deformation-response structure. The present paper shows that removing localized particle-class excitations does not leave nothing. It leaves the minimal concrete persistence baseline required for field-carrier mediation, boundary trace, quotient comparison, bounded transport, gauge compatibility, restoration, and deformation response. That baseline is the vacuum role.

What has been isolated is the minimal non-null field-carrier persistence regime left when localized particle-class excitations are absent. Familiar vacuum structures in QFT, zero-point physics, symmetry breaking, condensates, cosmological terms, and dark-energy models become later specializations of this role. They do not replace it. Vacuum is therefore not the null regime under another name; it is the minimal concrete persistence regime relative to which particles, fields, gauge structure, mass/inertia, and later dynamics are physically meaningful.

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